9th Class • Punjab Board • PCTB New Syllabus

Chapter 2: Biodiversity

Solved notes in book order — MCQs, short answers, detailed answers, inquisitive questions and key points.

Chapter 2 • Units 2.1 – 2.8 • Exercise solved in book order

What this chapter is about

• Biodiversity means the variety of living organisms in different ecosystems. It keeps ecosystems balanced, controls climate and gives us food, medicine, wood and fuel.

• Classification is the process of dividing organisms into groups and subgroups on the basis of their similarities and differences; the groups are called taxonomic ranks or taxa.

• The chapter follows the history of classification from Aristotle to Carl Woese, explains the three domains and the four kingdoms of Eukarya, the place of viruses, and the rules of binomial nomenclature.

• The textbook exercise (MCQs, short answers, detailed answers and inquisitive questions) is solved here in the same order as in the book.

A. Multiple Choice Questions

Tick the correct option. Read the reason given after each answer; it also tells you why the other options are wrong.

1. Which of the following taxonomic ranks represents the broadest rank?

a) Species
b) Genus
c) Kingdom
d) Domain

Answer: d) Domain — Domain is the highest (broadest) taxonomic rank; species is the lowest. All organisms are divided into three domains: Archaea, Bacteria and Eukarya.

2. Which characteristic is unique to organisms in the domain Archaea?

a) Cell walls made of peptidoglycan
b) Presence of a nucleus
c) Ability to live in extreme environments
d) Lack of ribosomes

Answer: c) Ability to live in extreme environments — The unique lipids of the archaeal cell membrane let them live in hot springs, salt lakes and acidic or alkaline waters; peptidoglycan is found in bacteria, not in archaea.

3. Which of these statements is NOT related to bacteria?

a) Do not have a nucleus
b) Cell wall made of peptidoglycan
c) Most are heterotrophic
d) Have chlorophyll in their chloroplast

Answer: d) Have chlorophyll in their chloroplast — Bacteria are prokaryotes, so they have no membrane-bound organelles such as chloroplasts; some of them do

have chlorophyll, but it is not inside a chloroplast.

4. Which of these organisms belongs to the domain Eukarya?

a) Escherichia coli
b) Yeast
c) Coronavirus
d) None of these

Answer: b) Yeast — Yeast is a fungus, so it is a eukaryote with a nucleus. Escherichia coli is a bacterium (domain Bacteria) and coronavirus is a virus, which is placed in no domain at all.

5. Which of the following is a key characteristic that distinguishes eukaryotic cells from prokaryotic cells?

a) Lack of a cell wall
b) Presence of a nucleus
c) Absence of ribosomes
d) Smaller size

Answer: b) Presence of a nucleus — Eukaryotic cells have a true nucleus and membrane-bound organelles; prokaryotic cells have no nuclear envelope. Ribosomes are present in both.

6. Which kingdom includes organisms that are primarily unicellular, eukaryotic, and often heterotrophic?

a) Archaea
b) Protista
c) Fungi
d) Plantae

Answer: b) Protista — Kingdom Protista contains unicellular or simple multicellular eukaryotes; most protozoans are heterotrophic. Archaea are prokaryotic and fungi are mostly multicellular.

7. Why are fungi included in heterotrophic organisms?

a) Have chitin in cell wall
b) Absorb nutrients
c) Reproduce by spores
d) Cannot prepare food

Answer: d) Cannot prepare food — Heterotroph means an organism that cannot make its own food. Fungi have no chlorophyll, so they absorb food from decaying matter; chitin and spores are not the reason of being heterotrophic.

8. Why it is impossible to classify viruses within traditional biological kingdoms?

a) They lack cellular structure and organelles
b) They cannot perform photosynthesis
c) They are smaller in size than bacteria
d) They are parasites

Answer: a) They lack cellular structure and organelles — All kingdoms are made of cells, but a virus is acellular — only a nucleic acid core with a protein coat — so it has none of the characteristics of the three domains of life.

9. Which of the following is the correct way for writing the scientific name of humans?

a) Homo sapiens
b) Homo Sapiens
c) Homo Sapiens
d) homo sapiens

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Answer: a) Homo sapiens — The genus name starts with a capital letter, the species name starts with a small letter, and in print the name is written in italics (underlined when handwritten).

10. Which information you can get if you know the scientific name of an organism?

a) Kingdom and phylum
b) Phylum and genus
c) Genus and species
d) Class and species

Answer: c) Genus and species — A binomial name has exactly two parts: the first part is the genus and the second part is the species of that organism.

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B. Write Short Answers

1. What is the term used to describe the variety of organisms in ecosystems?

Answer: Biodiversity

Biodiversity is the measure of the variety of living organisms present in different ecosystems, including terrestrial, marine and desert ecosystems. It includes the diversity of genes, of species and of ecosystems.

2. How is the biodiversity crucial for humans and for the planet Earth?

1. Ecosystem stability: biodiversity helps to maintain the balance of ecosystems and plays an important role in biogeochemical cycles such as the carbon cycle and the nitrogen cycle.

2. Climate regulation: plants and algae absorb carbon dioxide, which helps to keep the climate balanced.

3. Natural resources: biodiversity provides a vast array of resources, from food and medicine to building materials and fuel.

4. Economic benefits: biodiversity supports different industries, including agriculture, tourism and pharmaceuticals.

It also has aesthetic and moral value: many people respect biodiversity on aesthetic and moral grounds, not only because it is useful. That is why the United Nations has designated May 22 as International Biodiversity Day.

3. What are the seven taxonomic ranks used in the Linnaean system?

The seven ranks used by Carl Linnaeus (1735) are: **kingdom, phylum** (division in plants and fungi), **class, order, family, genus** and **species**. In 1977 the rank of **domain** was added above kingdom, so now there are eight ranks in all.

4. Write the taxonomic ranks of lion and corn.

Rank	Lion	Corn (Maize)
Domain	Eukarya	Eukarya
Kingdom	Animalia	Plantae
Phylum / Division	Chordata	Magnoliophyta (Angiospermophyta)
Class	Mammalia	Liliopsida (Monocots)
Order	Carnivora	Poales
Family	Felidae	Poaceae (Gramineae)
Genus	Panthera	Zea
Species	Panthera leo (lion)	Zea mays (corn / maize)

Table: Taxonomic ranks of lion and corn (compare it with the book's table of fruit fly, human and pea)

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5. What are the basic differences between archaea and bacteria?

Archaea	Bacteria
These are the most primitive (oldest) organisms on Earth.	These are also prokaryotes but they are not so ancient.
The cell wall does not contain peptidoglycan; it is made of various polypeptides and proteins.	The cell wall is made of peptidoglycan.
Their rRNA resembles eukaryotes more than bacteria, and the cell membrane contains unique lipids.	Their rRNA is typically bacterial and the membrane lipids are of the usual type.
They can live in extreme environments such as hot springs, salt lakes and acidic or alkaline waters.	They are found in all ordinary environments: soil, water, air and the bodies of organisms.
Some obtain energy from inorganic compounds such as sulphur or ammonia; some perform photosynthesis but do not produce oxygen.	Most are heterotrophic; some have chlorophyll and carry out photosynthesis.
Kingdom: Archaebacteria (e.g. methanogens, halophiles, thermophiles, acidophiles).	Kingdom: Eubacteria (true bacteria), e.g. decomposers and disease-causing bacteria.

6. What are the shortcomings of the three-kingdom classification system?

In 1866 the German zoologist Ernst Haeckel proposed the third kingdom Protista for Euglena-like organisms, and placed prokaryotes (bacteria and archaea) also in kingdom Protista. Its shortcomings were:

1. Fungi were still kept in kingdom Plantae, although fungi are heterotrophs which get their food by absorption and have chitin in their cell walls, not cellulose. Many taxonomists disagreed with this.

2. Prokaryotes (bacteria and archaea) and eukaryotes (protists) were placed in the same kingdom, so the basic difference between prokaryotic and eukaryotic cells was not cleared.

3. Very different organisms such as bacteria and Euglena were put together in one kingdom, which made the study of their relationships difficult.

7. Which kingdom includes organisms that are multicellular and heterotrophic, and lack cell walls?

Answer: Kingdom Animalia

Animals are eukaryotic, multicellular and heterotrophic. Their cells have no cell wall. They develop from embryos, ingest food and digest it within their bodies.

Vertebrates and invertebrates are both included in kingdom Animalia, e.g. lion, fish, insect.

8. Enlist the distinguishing characteristics of fungi.

1. Fungi are eukaryotic and heterotrophic, and their cells have a cell wall made of chitin.
2. They do not ingest food; they **absorb** food from decaying matter present in their surroundings.
3. Most fungi are multicellular (mushrooms, rusts, smuts, moulds) while a few are unicellular (yeast).
4. They have no chlorophyll, so they cannot prepare their own food.
5. They are useful as well as harmful: some are used in the production of bread, cheese and beer, and penicillin is obtained from the fungus Penicillium, while many cause diseases of plants and animals.

9. List the three domains that encompass all living organisms.

The three domains are: **(1) Archaea** — the most primitive prokaryotes, e.g. methanogens, halophiles, thermophiles; **(2) Bacteria** — prokaryotes with peptidoglycan in the cell wall, e.g. Eubacteria; and **(3) Eukarya** — all unicellular and multicellular eukaryotes, which contain the kingdoms Protista, Fungi, Plantae and Animalia.

10. Why cannot we classify viruses in any kingdom?

Viruses are ultramicroscopic creatures that stand at the borderline of living and non-living. They are **acellular**: they are not made of cells and have no organelles. A virus is only a nucleic acid (DNA or RNA) surrounded by a protein coat, and it cannot run metabolism of its own. To make its proteins and to increase in number it has to become a parasite inside a living organism. Because a virus lacks all the characteristics of the three domains of life (cell structure, nucleus, organelles, metabolism), it cannot be placed in any kingdom. Prions (protein only) and viroids (circular RNA only) are also acellular and are not included in the classification system.

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11. How does binomial nomenclature facilitate clear communication about organisms across different languages?

Common names differ from region to region and from language to language. For example, onion is 'Piyaz' in Urdu but in different regions of Pakistan it is also called 'ganda', 'bassal' or 'vassal', and other countries have other names for it. Sometimes one common name is used for many different organisms — 'blue bell' is used for dozens of plants and 'black bird' for both crow and raven. Common names have no scientific basis either: 'silver fish', 'cray fish', 'jelly fish' and 'star fish' are not true fishes.
Binomial nomenclature removes all this confusion. Its words are taken from the Latin language, which is spoken by no country, so no country is favoured. The scientific

name of an organism is unique and is the same anywhere in the world. So whether a biologist is Chinese, Arabic, Spanish or English speaking, the name Homo sapiens means the same organism to all of them. In this way binomial nomenclature provides one standard language of communication for biologists all over the world.

Remember the difference

- Bacteria: prokaryotic, peptidoglycan cell wall, ordinary environments.
- Archaea: prokaryotic, no peptidoglycan, unique lipids, extreme environments.
- Protista: eukaryotic, unicellular or simple multicellular, no multicellular sex organs.
- Fungi: eukaryotic, chitin cell wall, absorb food (heterotrophic).

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C. Write Answers in Detail

1. Discuss biodiversity and its significance in maintaining the health of ecosystems.

Definition: biodiversity is the measure of the variety of living organisms present in different ecosystems including terrestrial, marine and desert ecosystems. It covers the variety of genes, species and ecosystems. Biologists have identified about 2 million kinds of organisms — 0.5 million kinds of plants and 1.5 million kinds of animals — and every year thousands of new kinds are discovered.

Significance in maintaining the health of ecosystems:

1. Ecosystem stability: biodiversity keeps the balance of ecosystems. Different organisms depend on one another for food, so if one kind is lost, the whole food web is disturbed. Biodiversity also keeps the biogeochemical cycles (carbon cycle, nitrogen cycle) running, because decomposers, plants and animals take part in these cycles.

2. Climate regulation: plants and algae absorb carbon dioxide from the air during photosynthesis. Forests and oceans therefore act as carbon sinks and keep the climate balanced.

3. Protection of soil and water: plant cover and the animals living in the soil prevent erosion, keep the soil fertile by adding humus and help in the purification of water.

4. Natural resources: biodiversity provides a vast array of resources — food, medicine, building materials and fuel — so human life depends on it.

5. Economic benefits: biodiversity supports industries such as agriculture, tourism and pharmaceuticals, and it gives employment to millions of people.

6. Aesthetic and moral value: people also respect biodiversity on aesthetic and moral grounds; a rich natural world is beautiful and it is our duty to protect it. The United Nations has designated May 22 as International Biodiversity Day to promote the protection of biodiversity.

Conclusion: biodiversity is therefore not only a list of species but the working system of nature. Its loss disturbs food chains, climate and human economy, so its conservation is necessary for the health of the planet and for human beings.

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2. Explain the importance of classification in biology and how it helps us understand the relationships between different organisms.

Definition: classification is the process in which organisms are divided into groups and subgroups on the basis of the similarities and differences found in them. Biologists classify organisms on the basis of their physical characteristics and, in recent times, with the help of genetics — they find genetic similarities and differences and use this information to know similarities and differences in structure and function.

Aims of classification: (i) to determine similarities and differences among organisms so that they can be studied easily, and (ii) to find the evolutionary relationships among organisms.

Importance (advantages) of classification:

1. It allows biologists to group similar organisms together, which makes it easier to identify and understand their characteristics, relationships and evolutionary history. It helps us understand the vast diversity of living organisms on Earth.

2. It provides a framework for studying and comparing different species; a scientist can compare two organisms rank by rank instead of comparing everything at once.

3. It explains the inter-relationship among various organisms and shows which organisms are closer to one another.

4. It helps in the identification of new species and in understanding their evolutionary relationships.

5. It provides a common language for biologists around the world, enabling effective communication in the study of organisms.

Conclusion: overall, classification is crucial for our understanding of the natural world and for the conservation and management of biological diversity. When organisms are placed together on the basis of real similarities, the taxonomic ranks themselves tell us their relationships — the closer two organisms are in the ranks (same genus), the closer is their relationship and the more recent is their common ancestor.

3. Describe the Linnaean system of classification in detail, stating the seven taxonomic ranks and their relationships.

Introduction: the Swedish botanist Carl Linnaeus (1707-1778) created a taxonomic hierarchy of organisms and devised the Linnaean system of taxonomic ranks in 1735. He also introduced binomial nomenclature, that is, giving every organism a two-part scientific name. In 1977 the rank of domain was added above kingdom.

The seven taxonomic ranks of Linnaeus:

1. Kingdom: the domain is further divided into kingdoms, e.g. the domain Eukarya is divided into four kingdoms — Animalia, Plantae, Fungi and Protista.

2. Phylum (division in plants and fungi): each kingdom is subdivided into related phyla or divisions.

3. Class: each phylum or division is divided into related classes.

4. Order: each class is further divided into related orders.

5. Family: each order is broken down into related families.

6. Genus: each family is divided into related genera (singular genus).

7. Species: it is the lowest level of classification. A species is a group of similar organisms that can interbreed and produce fertile offspring.

Rank	Fruit fly	Human	Pea
Domain	Eukarya	Eukarya	Eukarya
Kingdom	Animalia	Animalia	Plantae
Phylum / Division	Arthropoda	Chordata	Magnoliophyta
Class	Insecta	Mammalia	Magnoliopsida

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Rank	Fruit fly	Human	Pea
Order	Diptera	Primates	Fabales
Family	Drosophilidae	Hominidae	Fabaceae
Genus	Drosophila	Homo	Pisum
Species	Drosophila melanogaster	Homo sapiens	Pisum sativum

Table: Classification of fruit fly, human and pea (from the textbook)

Relationships between the ranks: the ranks go from the broadest (domain) to the narrowest (species). As we move down, the number of organisms in a rank decreases while their similarities increase. Two organisms of the same species are most closely related, and the more ranks two organisms share, the closer is their relationship. Ranks are not fixed in size — a genus may have one species or many species — but the order of ranks is always the same.

4. Compare and contrast the domains Archaea and Bacteria, focusing on their key characteristics.

Similarities: both Archaea and Bacteria are prokaryotic — their cells have no nucleus and no membrane-bound organelles; both are unicellular and both are found in soil and water. Both were once placed in a single kingdom (Monera) before Carl Woese separated them in 1977.

Key point	Domain Archaea	Domain Bacteria
Cell type	Prokaryotic	Prokaryotic
Cell wall	No peptidoglycan; made of various polypeptides and proteins	Made of peptidoglycan
Cell membrane	Contains unique lipids that resist extreme conditions	Contains ordinary membrane lipids
rRNA	Resembles eukaryotes more than bacteria	Typical bacterial rRNA
Habitat	Extreme environments (hot springs, salt lakes, acidic or alkaline waters), also soil and oceans	All ordinary environments: soil, water, air and bodies of organisms
Mode of nutrition	Some get energy from inorganic compounds such as sulphur or ammonia; some photosynthesise but do not produce oxygen	Most are heterotrophic; some have chlorophyll and photosynthesise

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Key point	Domain Archaea	Domain Bacteria
Kingdom	Archaebacteria (methanogens, halophiles, thermophiles, acidophiles)	Eubacteria (true bacteria); many are decomposers, some cause diseases
Relation with disease	Not known to cause human disease	Many cause diseases of humans, animals and plants

Table: Comparison of domain Archaea and domain Bacteria

Conclusion: although both are prokaryotes, archaea are the most primitive organisms and differ from bacteria in cell wall, membrane lipids, rRNA and habitat. Their differences are so clear that they are placed in two separate domains.

5. Describe the diagnostic characteristics of the four kingdoms within the domain Eukarya.

Domain Eukarya includes all unicellular and multicellular eukaryotes. Their cells have a true nucleus and other membrane-bound organelles. It contains four kingdoms:

(a) Kingdom Protista: it includes eukaryotes which are unicellular, colonial or simple multicellular. Simple multicellular means that they do not have multicellular sex organs. There are three types: **plant-like protists (algae)** — cell wall of cellulose, chlorophyll in chloroplasts, autotrophic, e.g. Euglena, diatoms; **animal-like protists (protozoans)** — heterotrophic, ingest food, no cell wall, e.g. Amoeba, Paramecium; and **fungus-like protists (moulds)** — absorb nutrients from decaying organic matter, cell wall of cellulose instead of chitin, e.g. slime moulds and water moulds. Some protists are parasitic: Plasmodium causes malaria and Entamoeba causes amoebic dysentery.

(b) Kingdom Fungi: eukaryotic heterotrophs with a cell wall of chitin. They do not ingest food like animals; they absorb food from decaying matter. Most are multicellular (mushrooms, rusts, smuts, moulds) and a few are unicellular (yeast). They are useful in the production of bread, cheese and beer, and penicillin is obtained from Penicillium.

(c) Kingdom Plantae: eukaryotic, multicellular organisms with cell walls of cellulose. They are autotrophic and prepare food by photosynthesis. All plants have multicellular sex organs; during sexual reproduction they form embryos, and asexual reproduction through vegetative parts is also common. Examples: mosses, ferns, conifers and flowering plants.

(d) Kingdom Animalia: eukaryotic, multicellular and heterotrophic. They develop from embryos, ingest food and digest it within their bodies. Their cells have no cell wall. Vertebrates and invertebrates are both included, e.g. lion, fish, insects.

Feature	Protista	Fungi	Plantae	Animalia
Cell type	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic

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Feature	Protista	Fungi	Plantae	Animalia
Cell wall	Present in some, of various types	Chitin	Cellulose and other polysaccharides	Absent
Nutrition	Autotroph or heterotroph, or combination	Absorptive heterotroph	Autotroph	Ingestive heterotroph
Multi-cellularity	Absent in most	Present in most	Present in all	Present in all
Example	Euglena, Amoeba, Paramecium	Mushroom, yeast, Penicillium	Moss, fern, conifer, rose	Lion, fish, insect

Table: Characteristics of the domains and kingdoms of life (textbook table)

6. Discuss the challenges of classifying viruses within the traditional three domains of life.

What a virus is: viruses are ultramicroscopic creatures that lie at the borderline of living and non-living. They are acellular, that is, they are not made of cells and have no organelles. A virus consists of nucleic acid (DNA or RNA) surrounded by a protein coat. Common examples are the bacteriophage (a virus which attacks bacteria) and the influenza virus.

Challenges in classification:

1. No cellular structure: all three domains are defined on the basis of cell type (prokaryotic or eukaryotic), presence of a nuclear envelope and organelles. A virus has none of these, so there is no place for it in any domain or kingdom.

2. No independent metabolism: a virus cannot run its own metabolism. It becomes a parasite inside a living organism (plant, animal or bacterium) to make its proteins and to increase in number. So it cannot be classified by its mode of nutrition like other organisms.

3. No reproduction of its own: viruses do not reproduce by themselves; they multiply only by using the machinery of a host cell. Reproduction is a basic characteristic of living things, which makes the position of viruses doubtful.

4. Non-living like crystals: outside the host cell a virus is inert like a chemical crystal, but inside the host it behaves like a living thing. This dual nature makes a single clear decision impossible.

5. Prions and viroids: prions (made of protein only) and viroids (made of circular RNA only) are also acellular particles. They are not included in the classification system, yet both cause infectious diseases in certain plants and animals and are also a cause of cancer. So the list of acellular disease agents keeps growing and the three-domain system cannot adjust them.

6. Fast mutation and new viruses: new viruses appear again and again, such as coronavirus, identified in late 2019 in Wuhan, China, which caused the pandemic of

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respiratory illness called COVID-19. Such new viruses have to be studied and grouped separately (by their nucleic acid, shape and host) rather than by the ranks of the three domains.

Conclusion: hence viruses are not included in the classification system because they lack any of the characteristics of the three domains of life; they are studied in a separate group of their own.

7. Explain the rules and guidelines for suggesting scientific names to organisms.

Binomial nomenclature is the system of giving scientific names to organisms, started by the great Swedish biologist Carl Linnaeus. The scientific name of an organism consists of two parts: the first part is the name of the genus to which the organism belongs and the second part is the name of its species, e.g. Homo sapiens, Panthera tigris, Zea mays.

Rules of binomial nomenclature: the scientific naming of an organism is done in accordance with some international rules. Some important rules are:

1. For scientific naming, words are taken from the Latin language.

2. Every scientific name has two parts. The genus name always comes first, followed by the species name.

3. Every scientific name should be unique, because the same name cannot be used for naming two different organisms.

4. The first part of the name, i.e. the genus name, should begin with a capital letter, and the second part, i.e. the species name, should begin with a small letter.

5. At the time of printing, a scientific name should be typed in italics.

6. When a scientific name is hand written, the two parts of it should be separately underlined.

Significance of these rules: two organisms cannot have the same name; the words are from a language spoken by no country, so no country is favoured; and the name of an organism is the same anywhere in the world. This provides a standard way of communication whether the biologist is Chinese, Arabic, Spanish or English. Without such rules, common names would keep on confusing us — 'black bird' for both crow and raven, or 'jelly fish' for an animal that is not a fish.

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D. Inquisitive Questions

1. How might placing an organism in the incorrect taxonomic group affect conservation or scientific studies?

1. Wrong conservation priority: if a rare animal is placed in a wrong group, it may be counted as a common species and no protection programme will be started for it. On the other hand, money and resources may be spent on an organism that is not actually endangered.

2. Wrong protection laws: conservation laws and red lists are made group-wise, so a misclassified species may not come under the law that protects its real relatives, and hunters or traders may exploit it legally.

3. Wrong breeding and reintroduction plans: conservationists try to breed closely related organisms or to shift a species to a suitable habitat. If the relationship is wrong, the breeding pairs may be infertile, or the new habitat may not suit the organism at all.

4. Wrong medicine and agriculture decisions: if a fungus or a bacterium is placed in the wrong group, a wrong antibiotic, fungicide or crop-rotation plan may be chosen and the disease will not be controlled.

5. Wrong scientific conclusions: all research data, experiments and comparisons are organised by taxonomic groups. A wrong group gives wrong comparisons, wrong evolutionary trees and wrong conclusions, and other scientists waste time following that mistake.

6. Confusion in communication: binomial names are the common language of biology. A wrong name means different scientists talk about different organisms under the same name, and published research cannot be repeated or verified.

Conclusion: classification is not a formality; it decides how we study, protect and use an organism. That is why modern taxonomists check genetics along with structure before placing an organism in a group.

2. Imagine you discover a new organism. What steps would you take to classify and name it according to the principles of binomial nomenclature?

1. Careful study and observation: I would observe the organism in its habitat and note its structure, size, colour, cell type, mode of nutrition, mode of reproduction and behaviour. A microscope or a culture would be used if it is microscopic.

2. Decide the domain and kingdom: first of all I would see whether its cells are prokaryotic or eukaryotic. If the cells have no nucleus, it goes to Archaea or Bacteria (looking at the cell wall and habitat); if it has a nucleus and organelles, it goes to Eukarya — then, on the basis of cell wall, nutrition and multicellularity, it is placed in Protista, Fungi, Plantae or Animalia.

3. Fix the lower ranks: I would compare it with known organisms and fix its phylum (division), class, order, family, genus and species, using the Linnaean hierarchy and the keys and literature of taxonomy.

4. Confirm with genetics: in recent times taxonomists also take help from genetics. I would compare its DNA or rRNA with that of related organisms to confirm the position and to find its evolutionary relationships.

5. Give the scientific name: I would name it according to the rules of binomial nomenclature — two Latin words, the genus name first (starting with a capital letter) and the species name after it (starting with a small letter), the name printed in italics or underlined when handwritten, and a name that no other organism already has. For example, if it belongs to the genus Panthera and is a new species, its name would be written as Panthera (new word).

6. Publish and verify: I would write a scientific report with the description, photographs, measurements and references, and publish it in a scientific journal so that other taxonomists can check it. Only when the new name is accepted

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internationally does the organism become part of the classification system.

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Points to Remember

Biodiversity is the measure of the variety of living organisms present in different ecosystems including terrestrial, marine and desert ecosystems.

Importance of biodiversity: ecosystem stability and biogeochemical cycles, climate regulation, natural resources, economic benefits, and aesthetic and moral value. May 22 is International Biodiversity Day.

Classification is the process in which organisms are divided into groups and subgroups on the basis of the similarities and differences found in them. Its aims are to determine similarities and differences and to find evolutionary relationships.

Taxonomic ranks (taxa) are the groups of classification. Linnaeus (1735) gave seven ranks: kingdom, phylum, class, order, family, genus and species; the domain was added above kingdom in 1977.

History of classification: Aristotle — first divided organisms into plants and animals; Abu Usman Umer Al-Jahiz — described 350 species of animals; Andrea Caesalpinia — divided plants into fifteen groups calling them genera; Tournefort — introduced the taxa of class and species; Carl Linnaeus — created the seven-rank hierarchy, the two kingdoms and binomial nomenclature.

Systems of classification: two-kingdom (Plantae, Animalia); three-kingdom (Ernst Haeckel, 1866, added Protista); five-kingdom (E. Chatton gave the terms prokaryotic and eukaryotic in 1937; Robert Whittaker, 1969 — Monera, Protista, Fungi, Plantae, Animalia; Margulis and Schwartz modified it in 1988); three-domain (Carl Woese, 1977 — Archaea, Bacteria, Eukarya).

Domain Archaea: most primitive prokaryotes; no peptidoglycan in the cell wall; unique membrane lipids; live in extreme environments (hot springs, salt lakes, acidic and alkaline waters); kingdom Archaebacteria — methanogens, halophiles, thermophiles, acidophiles.

Domain Bacteria: prokaryotes with a peptidoglycan cell wall; found in all environments; mostly heterotrophic, some have chlorophyll; kingdom Eubacteria.

Domain Eukarya: all unicellular and multicellular eukaryotes with a nucleus and organelles; kingdoms Protista, Fungi, Plantae and Animalia.

Viruses are acellular (nucleic acid plus protein coat), cannot run metabolism and are parasites inside living cells, so they are not included in any kingdom. Prions (protein only) and viroids (RNA only) are also acellular and unclassified.

Binomial nomenclature was founded by Carl Linnaeus. The scientific name of every organism has two parts: first the name of the genus and second the name of its species. The words are Latin, the genus starts with a capital letter, the species with a small letter, and the name is printed in italics or underlined when handwritten.

System	Given by	Kingdoms included
Two-kingdom	Earliest system (Aristotle / Linnaeus)	Plantae, Animalia

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System	Given by	Kingdoms included
Three-kingdom	Ernst Haeckel, 1866	Protista, Plantae, Animalia
Five-kingdom	Robert Whittaker, 1969 (modified by Margulis and Schwartz, 1988)	Monera, Protista, Fungi, Plantae, Animalia
Three-domain (six kingdoms)	Carl Woese, 1977	Archaea (Archaebacteria), Bacteria (Eubacteria), Eukarya (Protista, Fungi, Plantae, Animalia)

Table: Classification systems in one look

How to attempt these questions in the paper

• MCQ: write only the correct option (a, b, c or d).

• Short question: answer in 2 to 4 sentences; for a definition give the definition and one example.

• Difference question: always draw a two-column table with proper headings.

• Detailed answer: start with a definition, then numbered points, and end with a conclusion line. Names with years (Haeckel 1866, Whittaker 1969, Woese 1977) carry marks.

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